

Execution-Grounded Obligation Runtime for Intermittent Pre-Disaster Geohazard Monitoring: Local Evidence, Deadline-Aligned Expiry, and Honest Infeasibility

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Abstract

After a communication loss, issued configurations and cached records keep producing consequences: a sampling regime still drains a battery it can no longer be told to relax, and records already past their usefulness keep competing for the few recovery opportunities. We ask *on what evidence, and at which layer, an execution runtime may stop those consequences* for a pre-disaster landslide deployment—battery-plus-solar LoRaWAN Class A nodes, an autonomous gateway, a 16 h cellular outage, and a sparse BeiDou short-message backup—with an agent runtime placed *above* the communication entities. We make three points. First, under partial observability the runtime must separate *fulfilment state* from *failure cause*. When the delayed mission table reaches the gateway, evidence legally held by then identifies the failure segment for only 53.3% of already-missed upgraded obligations, with zero errors on the labelled subset; the remaining 46.7% are intrinsically *unknown* (never sampled vs. sampled but not yet heard), and complete 100% segment attribution is available only as offline diagnosis. Second, complete diagnosis is not a precondition for correct local action. A node knows when its own records pass their business deadline and can stop worthless retransmission without resolving why the remote side is silent. This disposition is ordinary per-record expiration (a Bundle-Protocol lifetime aligned to the obligation’s delivery deadline): we show it is bit-identical to a prior “certificate-driven purge”, and that its basis and placement, not a new discard algorithm, carry the gain—outage on-time delivery 202→434 at the main point, a paired ten-seed +4.21 service points (95% CI +3.64 to +4.79, positive on all seeds), 2.52-fold outage delivery, expired backup records 2538→1, and node deaths 12→0; withholding expired *sends* at the gateway without releasing the source cache instead moves cost to the access segment (−8). Third, enforcement follows a locality principle—a constraint acts where evidence, authority, and the observe-decide-act delay jointly permit; a node-local night-energy floor removes deaths in the tested seeds while a centre envelope alone cannot, and a pre-provisioned local rhythm matches every centre-compiled service level for the tested static missions (a scoped negative result, not a global optimality claim). A real-LLM three-arm study is formative: it reveals that access receipts were mistaken for back-haul state in both directions, and an unknown-aware projector removes the healthy-link false alarms in offline replay; end-to-end and cross-model validation is specified as future work. The contribution is an execution-grounded obligation runtime that acts on locally certain evidence while never upgrading “unconfirmed” to “unreachable”.

1 Introduction

Where the agent sits. Prior emergency-communication research combines and recovers *given* network roles and infrastructure: it places UAVs, allocates bandwidth, re-plans topology, or sequences infrastructure-recovery tools. We instead place an agent runtime *above* these entities and

study its ability to sustain task execution as the entities themselves appear, degrade, disconnect, and recover. The setting is *pre-disaster* monitoring in a mountainous geohazard area, under tight energy and intermittent access/backhaul. The reference deployment keeps field nodes, a gateway with a cache and pre-provisioned autonomous rules, and intermittent backhaul; LoRaWAN Class A is the studied access (a field study used a custom LoRa link). Relay and backup backhaul enter only where hardware and standards support them. The agent never judges landslide risk.

What we expected, and what we found. An early hypothesis was that an agent jointly configuring collection and backup sending would deliver more than fixed rules. Symmetric tests against deliberately strong ordinary baselines (energy-aware sampling, obligation-driven max-coverage backup packing, piecewise mission views, FIFO repair, rolling MPC, and an omniscient offline packer) found no realisable on-time delivery gain from re-optimising collection, scheduling, or packing under fixed resources and the standard hold-until-acknowledged edge discipline, over the policy families we tested. We report this as a scoped negative result about the tested strategies, not as a proof that no better scheduler exists. Closing that throughput question did not close the research problem.

The problem that remains: consequences continue after contact is lost. Instrumenting a real LLM controller exposed a sharper failure than low throughput: the runtime could not tell *where* the mission broke, and it acted on that confusion in both directions. During a 16 h backhaul outage it read “nodes are heard over LoRa” as “backhaul is up” and kept emitting an elevated 300 s configuration the outage made unapplicable; on a healthy link a certificate variant read “not every node was heard this window” as “the control plane is not delivering”. The same aggregate telemetry (age-of-information, state-of-charge, a count of heard nodes) cannot distinguish data never sampled, data sitting at a node, data at the gateway, or a missing return opportunity, and cannot distinguish a command queued, in transit, or applied. Two effects then persist with no one to stop them: a configuration already issued keeps driving energy use through the night, and records already past their deadline keep filling the limited access batches.

Reframed question and core insight. The controller should compile a changing, authorised monitoring requirement into a plan that can actually be executed over sampling, access, energy, and return opportunities; when part cannot be honoured it should report the obligation, interval, and segment as one of three rigorously separate statements—**confirmed**, **unconfirmed**, and **unreachable**—rather than silently under-delivering or killing nodes to try. Our central finding is that *complete diagnosis is not required for correct local action*. The centre may be unable to distinguish a browned-out node from a lost link, yet the node itself knows exactly when its own records pass their business deadline and can stop retransmitting them; the evidence needed to stop a worthless local action can be strictly smaller than the evidence needed to diagnose the whole chain. This turns the runtime’s goal from “be certain everywhere” to “match the statement and the action to the evidence actually available at the acting layer”.

Contributions.

1. **Obligation chains under partial observability (§3,§5).** A per-obligation execution chain and a three-class statement semantics that separates fulfilment state from failure cause and permits an explicit *unknown*. At gateway arrival, locally legal evidence labels 53.3% of missed upgraded obligations with 100% precision on the labelled subset and leaves 46.7% unknown; time-aware offline attribution reaches 100%, correcting a prior mis-attribution of late access

to backhaul capacity. Compile-time no-return geometry is retained as a *conditional* structural certificate.

2. **Deadline-aligned expiry as a local release primitive (§6).** Releasing a record when its business delivery deadline passes is standard per-record expiration (BPv7 lifetime), and we prove a prior certificate-driven purge is bit-identical to it on ten seeds. The defensible novelty is the obligation-deadline basis (not packet age or hop TTL), the zero-downlink source placement, and cross-segment consistency: suppressing expired sends at the gateway alone back-pressures access (−8 on-time), whereas source-local expiry raises outage on-time delivery 2.52-fold with expired sends driven to zero.
3. **Execution locality and a layered runtime (§7).** A locality criterion—act where evidence is available, the action is authorised, and the observe-decide-act delay is shorter than the remaining intervention window—assigns a node-local night-energy floor, gateway-local packing and attribution, and a centre runtime restricted to what local autonomy cannot obtain. Deterministic tests show the floor removes deaths in the tested seeds while a centre envelope alone cannot; a pre-provisioned local rhythm matches centre-compiled service for the tested static missions. We state these as empirical, point-specific results.
4. **A formative real-agent study and an unknown-aware mechanism (§8.5).** A three-arm live LLM study with symmetric accounting shows that access receipts were used as backhaul state in both an optimistic and a pessimistic direction, that apparent honesty differences mixed in output-format reliability and an expert prompt packet, and that the prior “infeasibility log” count was prompt-taught wording. An unknown-aware v5 projector removes the healthy-link false assertions in offline replay and replaces a blanket night gate with a computed energy ledger; end-to-end v5 and cross-model runs are specified as future work.

We do *not* claim an LLM out-optimises a solver with the full task model; the deterministic projector, expiry rule, packer, and energy ledger are in-layer tools available to rules and agents alike, and rules benefiting from them is a system success rather than a failure.

2 Background, scenario, and standards grounding

Deployment. Fourteen sensor nodes in two groups monitor displacement (and rainfall) on a slope. Nodes sample and upload to a local gateway over LoRaWAN Class A; the gateway forwards to a central server over cellular backhaul, with a BeiDou short-message service as a sparse backup. Nodes are battery plus small solar. The mission is *pre-disaster* and routine, with authorised warning-level changes; the agent never judges landslide risk. Node count, role mapping, and disturbance parameters follow the frozen Task v1.1 instance manifest; an earlier 16-node sketch is not used as a field fact.

Standards-grounded authority. We use two Chinese geological-hazard monitoring industry standards, DZ/T 0460–2023 and DZ/T 0450–2023, retrieved and clause-audited in the repository [1, 2]. Relevant clauses: §5.3.3 permits *remote dynamic adjustment of sampling and upload frequency by warning level* (no numeric levels are given, so the periods we use are declared research choices); §8.4.1 defines four levels (red/orange/yellow/blue) and §8.4.2 permits upgrade, downgrade, or termination after a conference; §5.3.7 calls for dual-mode communication under weak coverage in high-risk extreme weather—a conditional clause that appears once, not a blanket dual-channel mandate, and the communication-layer standard does not impose it; §5.3.6 gives endurance targets.

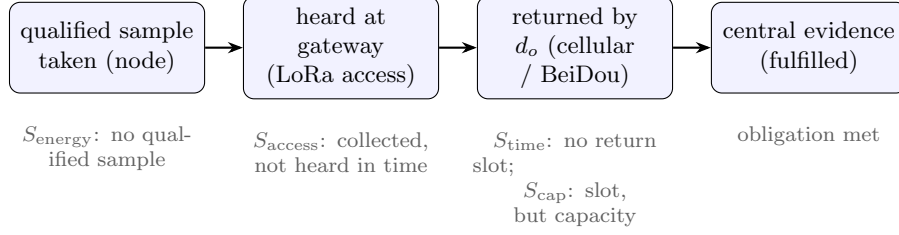


Figure 1: The per-obligation chain and the four failure segments. An unfulfilled obligation is charged to its *first* failing stage under an optimistic model that grants every later assumption, so the offline labels are mutually exclusive. An online observer that cannot see a stage must return *unknown* rather than guess it.

The BeiDou open-service specification (BDS-OS-PS-3.0) [3] reports $\geq 95\%$ success, average latency under 2 s, an average service frequency of one message per 30 s, and a single-message ceiling of 14,000 bits, while noting that realised frequency and maximum length are constrained by registration parameters—i.e. the rate is a *provisioned* quantity. These clauses bound what the runtime may lawfully do: re-compile frequency within granted levels and request a conference change, but never silently drop an obligation or declare risk.

Why the numbers are what they are. The short-message service is metered and sparse; the studied operating point uses one 78-byte packet every 1200 s (20 B header, 58 B payload; a displacement sample is 6 B and rainfall 4 B, so roughly 9–10 samples fit per packet). The elevated (yellow) task uses a 300 s sampling period with a delivery deadline two periods after window open, i.e. a ≈ 600 s delivery window—*shorter* than the 1200 s backup slot spacing. The 200 B single-packet ceiling and the ≥ 1 min period in DZ/T 0450 (§7.4.3.2, §7.4.2.3, with a ≥ 5 s inter-message gap at §7.4.3.5) belong to different service-formulation sections and are not combined into a single “BeiDou operating point”; the 1200 s/78 B point is a declared research choice.

3 System model and the obligation chain

Time is slotted at 60 s; $t=0$ is 06:00 local. A **monitoring obligation** o is a tuple $(n, m, [l_o, h_o), d_o)$ over node n , measurand m , sampling window $[l_o, h_o)$ and absolute delivery deadline d_o . A routine obligation at period P has $h_o - l_o = P$ and $d_o = h_o + P$ (one period of grace). We track a sample through a fixed chain:

$$\underbrace{\text{qualified sample taken}}_{\text{node}} \rightarrow \underbrace{\text{heard at gateway}}_{\text{LoRa access}} \rightarrow \underbrace{\text{returned before } d_o}_{\text{cellular / short-message}} \rightarrow \underbrace{\text{central evidence}}_{\text{mission fulfilled}}.$$

An obligation is *fulfilled* only when a sample matching (n, m) is taken in-window and received centrally by d_o . For piecewise missions the first segment uses a closed interval and later segments use half-open windows, so a boundary sample cannot double-cover adjacent windows after a level change. Figure 1 marks the chain stage at which each failure segment is charged.

Energy and devices. A sample costs 4.7×10^{-4} Wh; an uplink costs 2.33×10^{-5} Wh. Battery capacity is 0.05 Wh. Solar input follows a 12 h half-sine with peak 0.03 Wh/h, with per-node/hour stochastic cloud occlusion. A node that browns out *permanently* dies in the main configuration (an absorbing state; a restart sensitivity is left open). A cache of 240 slots with FIFO replacement

sits at the node. Under the modelled standard edge discipline a record leaves this cache only once it is acknowledged at the centre (the acknowledgement is modelled synchronously and without airtime, a pre-existing simplification that favours every queue discipline equally); until then it is retransmitted oldest-first, with at most 32 records per upload opportunity. Access succeeds with probability 0.74; the primary backhaul is good with probability 0.62.

Class A execution reality. A configuration command is not immediate: it is queued, transits the backhaul to the gateway, and reaches a node only in its next Class A receive window; it counts as APPLIED only when the node later reports the new config. During a backhaul outage, commands are refused at admission (they are never transmitted and occupy no airtime, so their count is a command-hygiene metric, not a wireless or short-message billing saving). A command issued now, applied later, and still consuming energy in between is a first-class effect; holding a dense 300 s regime across a 12 h night needs ≈ 0.071 Wh, exceeding the 0.05 Wh battery.

Mission reachability and who knows what, when. The authorised schedule is piecewise (blue 600 s, then yellow 300 s from $t=6$ h). The centre is pre-provisioned with the full authorised schedule; the *field gateway's* obligation view, however, is gated by a piecewise mission view and is published incrementally only as the mission table traverses the backhaul. In the main scenario the primary is down from $t=4$ h to $t=20$ h, and the table (which does not use the short-message path) reaches the gateway only at $t=20$ h; before then the field lawfully continues the last APPLIED blue configuration under local autonomy. We therefore do not model the field learning of an unseen future task at runtime; we model a pre-authorised schedule whose field view is delayed by the outage.

Three statement classes. Throughout, the runtime keeps distinct: **confirmed** (an independent node report shows the target config, or a matching sample is received centrally); **unconfirmed** (a command is in flight or receipts are incomplete—this implies neither that it took effect nor that the channel is down); and **unreachable** (the current path is known down from an explicit network-side state). A LoRa access receipt (“a node was heard”) is evidence about the access link only and can never by itself establish backhaul reachability.

Strong baselines (the bar to beat). We assemble deliberately strong ordinary mechanisms with the same legal information: **dayfeed**, an energy-aware policy that densifies by day and relaxes overnight; **maxcov**, a max-coverage/recency greedy that packs each backup packet by marginal new obligation coverage per byte; **piecewise mission views** with a gateway gate that withholds future requirements until they arrive while preserving already-open windows; and **FIFO** access repair. A scripted dayfeed decider driven through the full agent harness reproduces the reference service level bit-for-bit, demonstrating no hidden privilege or penalty for the agent path.

4 The feasibility frontier, scoped to the policies tested

Before proposing any mechanism we ask whether controllable, changeable loss exists, using *symmetric* per-obligation counterfactuals: each relaxation widens one physical resource (never the policy), and we report set differences over the same obligation ids, so a single strategy’s good luck cannot masquerade as an upper bound.

Failure-segment attribution (offline, time-aware). For every unfulfilled obligation we attribute the first blocking segment under an optimistic model that favours feasibility: S_{time} when even the optimistic model finds no primary or backup return opportunity between the qualified sample and d_o ; S_{energy} when an opportunity exists but no qualified sample is collected; S_{access} when a sample is collected but is not heard at the gateway *by the deadline*; and S_{cap} when it is heard in time but capacity/congestion cannot return it. Only the first failing segment is charged. Over the full 48 h, of 7560 routine obligations the strong maxcov/FIFO baseline delivers 3025 (0.400); the 4535 failures attribute offline as S_{energy} 2002 (26.5%), S_{cap} 1945 (25.7%), S_{time} 588 (7.8%), S_{access} 0. This full-horizon accounting is an offline diagnostic; the online-visible subset is quantified in §5 and is substantially smaller.

Relaxation walls over the tested policy family. Table 1 relaxes energy (battery 0.05→0.50 Wh with always-dense sampling) and backhaul (the 1200 s/78 B point to 60 s with unlimited bytes) independently and jointly. These relaxations change resources *and* couple in a specific policy, and

Run	Sampling/Energy	Backhaul	Delivered
base	dayfeed, 0.05 Wh	1200 s/78 B	3025 (0.400)
O_{bh}	dayfeed, 0.05 Wh	unlimited	4740 (0.627)
O_{en}	always-300, 0.50 Wh	1200 s/78 B	4089 (0.541)
O_{bo}	always-300, 0.50 Wh	unlimited	6023 (0.797)

Table 1: Counterfactual interventions over 7560 obligations. Set decomposition over the tested configurations: 1715 obligations are blocked by backhaul capacity alone, 1128 by battery alone, and 155 (2.0%) require both. These are intervention results on the implemented strategies, not an exhaustive optimum over all legal policies; we make no zero-remainder or global-optimality claim.

the four-segment labelling is an exhaustive partition of the implemented model rather than a proof that every residual is physically immutable; we therefore read Table 1 as evidence that, for the policy families we tested, the large fixed-resource gains require adding backhaul capacity or energy, not smarter collection/scheduling/packing. A later section identifies one fixed-resource gain that is orthogonal to all of these—a change to record *lifetime disposition* (§6), which was held constant here.

The sampling/energy policy family offers no robust operating point. A feed-forward night-energy policy that ignores two costs kills 13 nodes (service 0.213): it omits the Class A command latency L (the node keeps sampling densely while the relax command propagates) and the dawn charging ramp R (solar input is near zero at sunrise). Counting both ($L=3600$, $R=7200$ s) collapses the policy bit-for-bit to dayfeed. A rolling-horizon MPC that integrates the true half-sine harvest gains 1.47 points on seed 0 but averages -1.3 points over six seeds and introduces 21 new permanent deaths on seeds dayfeed survives; a robust sweep over the cloud coefficient has no operating point that adds no deaths and is not worse on average. Legal information does not reveal tomorrow’s clouds, while dense sampling under random cloud plus an absorbing brownout state has asymmetric cost. This bounds the specific MPC/threshold policies tested; it is not a claim that no energy controller can help under any formulation.

The outage window: mission unreachability dominates. Of the 2338 yellow obligations whose deadlines fall inside the outage ($\text{release} \geq 6$ h, $\text{deadline} \leq 20$ h), removing the energy wall gives zero gain (the yellow command cannot traverse the outage, so the field keeps sampling at blue 600 s regardless of battery), while widening the backhaul adds 918; 1218 remain unfulfilled even under both interventions (no-slot geometry plus access loss). An omniscient compile-time EDF packing that assumes every obligation’s sample is present at window start claims 420 obligations fit the backup slots, yet only 9 are realised—the offline packing was optimistic about *supply*, which the unreachable mission table prevents. Feasibility projection must take the field’s legally knowable supply rhythm as input, not the nominal mission table.

5 Feasibility statements under partial observability

The runtime emits obligation-level statements, kept rigorously distinct: **fulfilled** (verified by an independent scorer), **unconfirmed / not currently guaranteed** (a knowledge boundary), and **infeasible-under-stated-conditions** (an optimistic model must also fail; a heuristic that merely did not finish may never claim impossibility). Crucially, “not currently guaranteed” is never upgraded to “unreachable” without network-side evidence.

Two placement levels. At **compile time (centre)**, when an authorised change is authored, the centre holds the full obligation table, the network-side outage status (current state, not a forecast of its duration), and public device/link parameters. At **arrival time (gateway)**, when a possibly delayed mission table arrives, the gateway has the richest *local* state—samples it has actually heard, its own forwarding log, and backup slots used—but, by construction, nothing that has not yet reached it.

Conditional structural no-return certificate (S_{time}). Backup slots occur at $t \bmod 1200 = 0$; an upgraded obligation’s delivery window is 600 s. An obligation whose window contains no backup slot cannot be returned via backup even under optimistic supply and unlimited node energy; this is a geometric fact, seed-invariant. The certificate is *conditional*: it establishes S_{time} only **if the primary is in fact unavailable across that window**. Given the network-confirmed outage interval [4h, 20h], the test emits 602 no-slot certificates; 588 obligations truly end as S_{time} (100% recall of the 588, zero false positives), with 14 obligations at the recovery edge (20 h) that an optimistic primary slot could still serve and that are therefore excluded as boundary cases rather than certified. Lead time ranges 0.2–14 h with median 7.2 h, *conditional on the network-side unavailable interval being known*; geometry alone never asserts an outage. A capacity certificate is deliberately not issued at compile time because it is impure under our tests (552 correct S_{cap} , 571 actually S_{energy} , 193 false positives); capacity attribution is pushed to the gateway.

Arrival-time attribution: online partial, offline complete. When the delayed table arrives at $t_a = 20$ h, 2136 yellow obligations are already past deadline. We contrast two attributions (Table 2). The *offline, time-aware* diagnostic uses simulator-wide evidence—whether a sample was ever taken, when it was heard, and when it was returned—and labels all 2136 correctly. The *online gateway-local* attribution uses only evidence the gateway legally holds at t_a (samples heard or received by t_a) and returns an explicit unknown whenever the failure stage is not separable. The online-labelled subset is exact (1138/1138) but partial: 427 matching samples are first heard after t_a , and 650 after their deadline. The result sharpens the runtime’s requirement: the projector must

Segment	Offline time-aware truth	Online gateway-local at t_a
S_{time} (no return slot; geometry)	588	588
S_{cap} (heard by deadline, not returned)	327	327
S_{access} (heard after deadline / late)	650	223
S_{energy} (no qualified sample)	571	— (merged into unknown)
U (not separable online)	—	998
Total	2136	2136
Labelled coverage	100%	53.3% (1138/2136)
Precision on labelled subset	1.000	1.000

Table 2: Arrival-time attribution of the 2136 upgraded obligations already missed when their table reaches the gateway. Geometry needs no sample and is visible online; a sample heard by the deadline but not returned is provably S_{cap} to the gateway; a sample heard in $(\text{deadline}, t_a]$ is provably late access. The remaining 998 (46.7%) have no sample heard by t_a : the gateway cannot separate “never sampled” (S_{energy} , 571 in truth) from “sampled but heard only after t_a ” (S_{access} , 427 in truth), and must report unknown under the original G5 constraint. The earlier “100% arrival attribution” was the offline column; it also merged 650 late-access cases into a 977-strong S_{cap} class (977=327+650).

expose *unknown* as a first-class state, and local mechanisms must be able to act without resolving it.

6 The obligation runtime: local release and execution placement

Why withholding expired sends alone is not free. During the outage, expired backup records are the obvious reading of wasted metered capacity. But under the hold-until-acknowledged discipline, withholding a backup record also withholds its acknowledgement; the record stays in the node cache and, as an oldest unacknowledged entry, fills the limited access batches, so genuinely fresh samples are not heard until the primary recovers. At the main point, gateway-only suppression (stop sending expired records, leave the node cache untouched) saves backup bytes but costs 8 on-time obligations (202→194) and 0.25 service points: cost is moved from the backhaul to the access segment, not removed.

Deadline-aligned expiry. The execution-layer counterpart releases an unacknowledged edge record when its business delivery deadline has passed under the locally known period ($d \leq t$)—a zero-false-positive test because the deadline is absolute—while retaining every record with $d > t$. This both stops the expired backup send and frees the access batch. We implement and compare two formulations. The node-local **deadline-purge** needs no downlink at all; a gateway **cert-purge** locally acknowledges records it will not send. We further implement **generic-expiry**, an ordinary per-record expiration primitive: at record creation the source assigns a creation-relative lifetime ending at the business deadline, $\text{expires} = (\lfloor t_{\text{taken}}/P \rfloor + 2)P$, expired records are deleted and their retention released, unexpired records are kept, and survivors are sent FIFO. It references no obligation id, return-slot geometry, or certificate.

The “certificate” mechanism is standard expiry. Across all ten seeds, **deadline-purge** and **generic-expiry** produce bit-identical trajectories—identical service, outage on-time count, deaths, backup on-time/expired counts, and the same delivered-obligation set (Table 3). We therefore make no novel-discard-algorithm claim. Dropping stale or expired packets is well established in AoI

packet management and in DTN bundle expiry [4, 5, 6]; BPv7 attaches a per-bundle creation-time lifetime and deletes on expiry [6]. The system contribution is the *combination*: (i) the lifetime is set from the obligation’s end-to-end business delivery deadline rather than packet age, a hop-local TTL, or an estimated single-packet delay; (ii) release is executed at the source with zero downlink, on evidence the source holds unconditionally; and (iii) disposition is consistent across the access and backhaul segments, which the gateway-suppression experiment shows is necessary. Under a single fixed-period mission the obligation-keyed and deadline-keyed formulations coincide; when one record supports several overlapping obligations or mission versions, release must instead key to the *latest* deadline for which the record is still valid—a generalisation we leave to future work rather than claim.

Edge queue / action	Full svc (mean)	Outage on-time (10-seed total)	Expired backup records	Deaths
maxcov + FIFO (baseline)	0.373	1691	2538	12
maxcov + latest-only (AoI)	0.358	4250	0	0
maxcov + generic-expiry (standard)	0.415	4265	1	0
maxcov + deadline-purge (node)	0.415	4265	1	0

Table 3: Ten-seed sweep with the strong maxcov backup chooser fixed; only the edge disposition changes. The standard per-record expiration and the prior “certificate purge” are bit-identical on every seed. Expiry doubles outage on-time delivery while latest-only reaches outage freshness by abandoning post-recovery coverage (lower full service). FIFO retains expired records and incurs 12 deaths. The gateway-only suppression point (not shown; same FIFO edge cache) gives 194 outage on-time at seed 0 vs. 202 for FIFO and 434 for source expiry—the cross-segment cost.

Why ordinary reordering queues do not suffice. EDF and obligation-greedy disciplines reorder which records occupy a batch but never release them, so the cache is not freed and we measure no gain. The AoI latest-only queue does release records, but indiscriminately (it keeps only the newest), matching outage freshness while permanently abandoning records still deliverable after recovery and lowering full-horizon service. Deadline-aligned expiry supplies the *selectivity* that turns a blunt freshness heuristic into a service-preserving action; a fixed rule realises it, and an agent must maintain the same deadline judgement when requirements change.

The local-certainty principle. The expiry result evidences the paper’s central mechanism: at the moment $d \leq t$ the node holds a certain fact (its own record’s business deadline has passed) whose required action (stop retransmitting, free the batch) is entirely local, even though the node cannot know whether the remote silence is a brownout, an access loss, or a backhaul outage. Complete chain diagnosis is unnecessary for this disposition; demanding it would leave the worthless retransmission running.

7 Execution locality and the layered runtime

The source-expiry result hints at a placement principle; we make it explicit and test it deterministically (no LLM, three seeds, the main operating point). A feasibility constraint can be enforced in time at a layer only when three conditions jointly hold:

$$\text{evidence available} \wedge \text{action authorised} \wedge D_{\text{observe+decide+act}} < M_c(t),$$

where $M_c(t)$ is the remaining intervention window for constraint c . The question is not how smart a centre planner is, but where these three hold.

Two deterministic mechanisms. The first is a centre-side feasibility **envelope**, a pure filter wrapping any planner (rule, MPC, or LLM) and using only signals it may legally see (each node’s last *confirmed* configuration, recent node-hearing counts, the authorised requirement, and the local clock). It enforces a night-energy gate (18:00–06:00 sparse, dusk downgrade, 16:00–18:00 closed to new dense sessions), a control-plane confirmation gate, and sparse-under-blue. The second is a **node-local energy floor**: on every 60 s tick the node itself, on its local clock and reported SoC, clamps sampling/report to sparse at night and self-downgrades at dusk with no centre command; an optional pre-provisioned schedule additionally lets it execute the authorised day/night rhythm locally. This models the “pre-provisioned autonomous rules” already in the deployment; it is off by default and all pre-existing tests remain bit-identical when off. Unlike the v4 advisory, the floor’s night decision is a computed ledger (remaining dark hours $\times$ the dense-vs-sparse extra draw vs. reported SoC), not a blanket “dense kills” claim.

Why a centre envelope cannot ensure survival. A dense 300 s night costs ≈ 0.071 Wh, above the 0.05 Wh battery, and a brownout is absorbing (permanent within the model). A centre-issued dusk downgrade must traverse the cellular backhaul and then a Class A receive window; under the outage and geometric loss it does not arrive before dusk on the tightest-energy seed. The same rule—“never stay dense overnight”—is a late, refusable *suggestion* at the centre but an invariant enforced every tick *at the node*. Table 4 shows the consequence over the tested seeds.

Configuration (centre placement unless noted)	Mean svc	Deaths (3 seeds)	Cmds sent	Refused
comply (naïve, no protection)	0.162	39	160	2488
dayfeed-c (centre day/night rule)	0.370	12	308	1256
envelope(comply), centre only	0.359	12	308	298
comply + local floor	0.366	0	1424	3364
dayfeed-c + local floor	0.368	0	258	1156
envelope(comply) + local floor	0.359	0	282	222
local-dayfeed (existing conservative local policy)	0.296	0	0	0
pure-local (pre-provisioned, 0 centre cmd)	0.360	0	0	0

Table 4: Deterministic three-seed placement study (no LLM). The local floor removes all permanent deaths *in the three tested seeds* at unchanged service; the envelope removes the command storm (refused 2488→222) but cannot save the tightest seed without the floor. Refused commands are rejected at admission and never transmitted, so this is command hygiene, not a radio/short- message billing saving. *pure-local*—a node-local pre-provisioned rhythm with zero centre commands—matches every centre-compiled service level for the tested static missions. The slightly higher unprotected dayfeed-c mean is bought by 12 deaths and is not a survivable point. These are point-specific empirical results, not a general survival guarantee or an optimal-controller bound.

A local upper bound for static missions. *pure-local* attains mean service 0.360 (range 0.317–0.387) with zero deaths and zero centre commands, statistically indistinguishable from the centre-compiled configurations (0.359–0.368); a run in which the centre still issues comply commands while the node executes the pre-provisioned rhythm gives bit-identical service with 1274 sent / 2354 refused commands—centre sampling commands are pure overhead under a static, predictable mission. This closes the centre throughput/energy control question *for the tested static, pre-provisioned missions*; it does not bound non-pre-provisioned runtime task changes, which are the centre runtime’s actual remit.

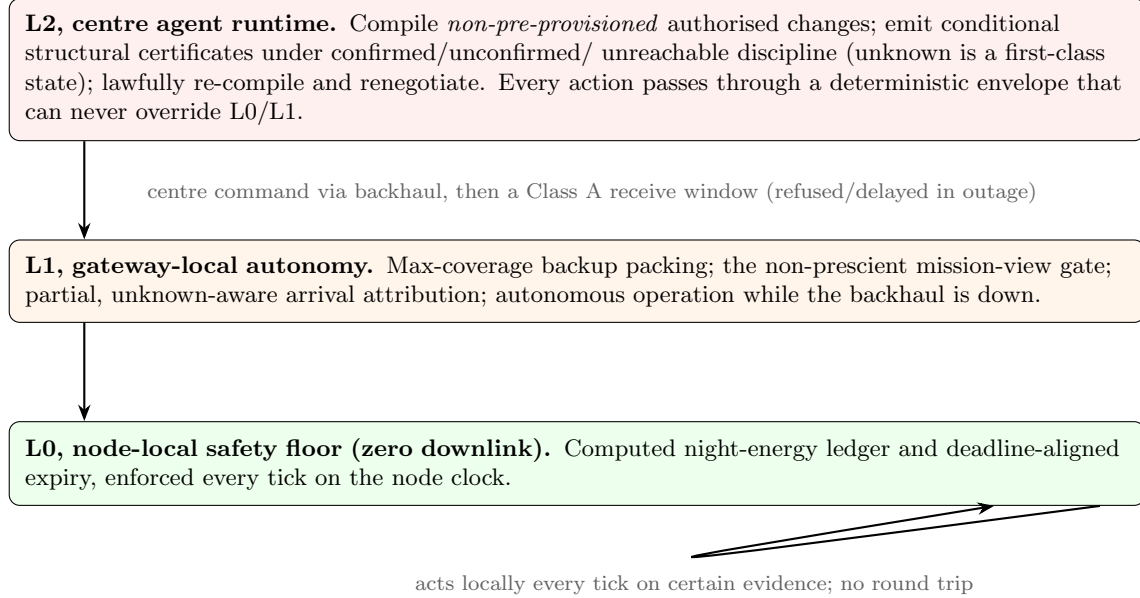


Figure 2: The layered obligation runtime. Safety-critical and deadline-certain constraints sit at L0, where they act before energy or access opportunity is spent without any round trip; L1 holds field-local packing and partial attribution; the L2 agent is confined to changes local autonomy cannot obtain, and must report unconfirmed rather than unreachable without network-side evidence.

The layered runtime. The results prescribe three layers, each holding only constraints it can act on in time (Figure 2): **L0, node-local (zero-signalling safety floor)**—the computed night-energy ledger and deadline-aligned expiry, acting every tick on the node clock to prevent irreversible discharge and dead-record cache exhaustion; **L1, gateway-local (field autonomy)**—max-coverage backup packing, the non-prescient mission-view gate, local (partial, unknown-aware) arrival attribution, and autonomous operation during outages; and **L2, centre agent runtime**—only what local autonomy cannot obtain: cross-layer compilation and lawful re-compilation/renegotiation of *non-pre-provisioned* authorised changes, compile-time conditional structural certificates under the three-way statement discipline, and honest external reporting. Every L2 action passes through the deterministic envelope and can never override L0/L1. Rules, MPC, and solvers benefit from the same layers: this is a general communication execution substrate, not a claim that an LLM is irreplaceable. Under static pre-provisioned missions L2 rightly stays silent.

8 Evaluation

Setup. Seed 0 main point (with paired multi-seed checks where randomness enters): 48 h plus a 1 h tail, 14 nodes in two groups, cap 0.05 Wh, solar peak 0.03 Wh/h, outage 4–20 h, backup 1200 s/78 B with maxcov, blue→yellow (600→300 s) at 6 h authorised at the centre with the field mission table reaching the gateway at 20 h. Access $p=0.74$, primary backhaul $p_{\text{good}}=0.62$. All numbers are produced by repository scripts; each run reproduces the repository checks and the five joint invariants; the local floor is off by default and the disabled-backup/no-floor configurations remain bit-identical to the v1.1 anchors. The placement study is deterministic over seeds 0–2; node death is a deterministic/highly robust quantity, while mean-service differences below one point are not claimed significant. Seeds 6–9 are a repeatedly reused held-out confirmation set, not an

untouched test set.

8.1 RQ1: Do structural certificates provide early, precise warning?

Conditionally, and deterministically. Given the network-confirmed outage interval, the geometry test emits 602 no-slot certificates; 588 obligations truly end as S_{time} (100% recall, zero false positives), the 14 recovery-edge cases are excluded rather than certified, and lead time is 0.2–14 h (median 7.2 h). Without a confirmed unavailable interval the same geometry is only a conditional statement (“unreturnable *if* the primary stays down”), which is how we now state it; geometry never infers an outage. An optimistic-energy capacity certificate is impure (552 correct S_{cap} , 571 actually S_{energy} , 193 false positives), so compile-time claims are restricted to structural/unreachable segments and capacity attribution is pushed to the gateway.

8.2 RQ2: What can the field attribute when the table finally arrives?

Partially online, completely offline (Table 2). Using only evidence held by $t_a=20$ h, the gateway labels 1138/2136 (53.3%) missed upgraded obligations—all geometric S_{time} (588), all in-time-but-not-returned S_{cap} (327), and 223 provably late S_{access} —with 100% precision on that labelled subset; 998 (46.7%) are unknown, whose offline truth is 571 never-sampled and 427 heard only after t_a . The offline time-aware diagnostic labels all 2136 correctly (588/327/650/571); it also corrects the earlier online figure, which had folded the 650 late-access cases into S_{cap} . The runtime value at this level is therefore not omniscient diagnosis but calibrated partial diagnosis plus an explicit unknown, which is exactly the state the local mechanisms of §6 operate despite.

8.3 RQ3: Does deadline-aligned expiry recover wasted cost?

During the outage 75 backup packets carry 750 records. With FIFO, of the records received centrally via backup during the outage a large fraction arrive already expired (219 expired vs. 261 on-time at seed 0, 45.6%; 1270 of 2782 payload bytes). Gateway-only suppression removes those sends but loses 8 on-time obligations (202→194) through access back-pressure. Source-local deadline expiry reverses the sign: at seed 0 it raises outage on-time obligations 202→434 and full-horizon service 0.400→0.432, drives expired backup records to zero, and raises the deadline-valid delivery fraction from 0.54 to 1.00; the gateway version is within 0.2 points (0.430/420), confirming the gain is the deadline judgement rather than its location. Across ten seeds (Table 3) expiry raises mean service 0.373→0.415—a paired +4.21 points (sd 0.81, 95% CI +3.64 to +4.79, positive on all ten seeds)—and total outage on-time delivery 1691→4265 (2.52×), with expired backup records 2538→1 and deaths 12→0; it dominates AoI latest-only on every seed by +5.73 points (95% CI +5.06 to +6.39), because latest-only matches outage freshness only by abandoning post-recovery coverage. EDF and obligation-greedy queues, which reorder but never release, gain nothing.

8.4 RQ4: Does enforcement placement determine safety and command hygiene?

Yes, deterministically, within the tested seeds (Table 4). (i) Naïve centre compliance permanently kills 13/14 nodes on all three tested seeds (39 total) and halves service to 0.162. (ii) A centre envelope removes the command storm (refused 2488→298) and protects two seeds but 12 nodes still die on the tightest seed because the dusk downgrade cannot cross the backhaul and a Class A window in time; the node-local floor removes all 39/12 deaths at unchanged service. (iii) With the floor present, adding the envelope cuts refused centre commands from 3364 (naïve)

or 1156 (centre dayfeed) to 222 at the same zero-death service level—a reduction in futile downlink *attempts* (rejected at admission, not transmitted). (iv) A zero-command pre-provisioned rhythm reaches 0.360 service, matching every centre-compiled arm for these static missions. We report zero deaths as an empirical result over the three tested seeds and operating point, not a universal survival invariant, and we do not transcribe these deterministic zeros onto the live agent runs of RQ5, which did not wire the L0 floor into the agent loop.

8.5 RQ5: What does a real agent show, and what does it not?

We ran a live three-arm study with deepseek-flash over the 16 h outage (three seeds for an unaided agent A0, the same agent with an ordinary pending-command checklist A0s, and a certificate agent A1) plus two healthy-link controls, each a full 48 h+tail run with a real model call at every 30 minute decision (99 decisions per trace). All arms share the harness, legal observation, tools, history, and action space; only the decider’s information block differs. We report the service and death figures, then restate the honesty/accounting claims under symmetric scoring.

Service and survival (robust counts; noisy service at $n = 3$). Table 5 reports per-seed service and permanent deaths. A1 attains the highest mean service (.358), statistically level with the deterministic safe floor (.360); seed-1 service overlaps across arms, so we draw no service significance from $n = 3$. Death counts are the more robust quantity.

Arm	svc (s0/s1/s2)	svc mean	deaths (s0/s1/s2; total)	dense orders (3 outage seeds)
A0 unaided	.160/.319/.408	.296	13/13/9 (35)	144
A0s + checklist	.181/.221/.265	.222	13/13/13 (39)	192
A1 + v4 certificate	.381/.316/.377	.358	0/4/0 (4)	67
A0 healthy-link control	.275		13	38
A1 healthy-link control	.511		0	28

Table 5: Live three-arm runs, 99 decisions per trace. The v0.6 “false reports / infeasibility log” columns are removed: under symmetric relabelling they did not measure what they claimed (see text). Dense orders count decisions that emitted a 300 s target. The four A1 seed-1 deaths are *not* cleared by L0 in this run, because the live agent loop was not wired to the L0 floor; the deterministic floor result of RQ4 is a separate experiment.

Accurate call accounting. The 11 traces contain 1089 decisions, not 1089 successful API calls. 52 decisions ended in a parse-failure hold, caused by `max_tokens` cutting the JSON mid-string (a second thinking-disabled attempt then also failed); 412 decision records carry a first-attempt error even when the retry succeeded (a logging artefact where the first error was not cleared). Per-arm parse-failure rates are A0 7.1% (28/396), A0s 6.4% (19/297), A1 1.3% (5/396): the arm difference therefore mixes output-format reliability with any semantic effect of the certificate. The hardened harness now logs per-request attempts, finish reasons, token use, and a truncation marker for future runs.

The honesty claims must be restated. The v0.6 scorer (i) scored a false “upgrade in effect” claim only inside decisions that ordered dense, so an arm that mostly holds (A1) was scored on the empty set; (ii) never ran on the healthy-link group (its outage window was set to none), yielding initialisation zeros; and (iii) counted a broad keyword set (hold/guarantee/certificate/downgrade)

that the certificate text itself taught the model. Symmetric relabelling of every decision with denominators, plus the interface fact that the observation’s `link` field carries only LoRa access receipts (`n_heard`) and *no backhaul state*, gives a different picture. In the forced outage window, A0/A0s assert “link up / commands can apply” on 36/32 decisions (A1 on 2)—a deterministic over-optimistic error, since the primary is down throughout; these are access receipts misread as a live backhaul, and they accompany the ineffective dense orders and deaths. Conversely, the v4 certificate asserts “control plane not delivering” whenever `heard < n`; offline replay on the healthy-link control confirms five decision times (local clock 14:00, 17:30 on day 1 and 07:00, 11:00, 06:30 on day 2) at which the primary was usable yet the assertion fired—twice while 13/14 nodes had already confirmed the 300 s regime—an over-pessimistic upgrade of *unconfirmed* to *unreachable*. The prior “89 structural infeasibility entries” are prompt-taught wording (the corresponding terms appear in 317 A1 decisions and in zero A0/A0s decisions), not per-obligation structural registrations. The honest finding is therefore not “the certificate makes the model truthful” but “a missing backhaul-state signal causes bidirectional errors, and the prompt packet changes both behaviour and wording”; isolating the certificate’s independent increment requires the matched-capability control below.

Unknown-aware v5 projector (offline mechanism witness). We repaired the projector without re-running the LLM: *unreachable* is emitted only from an explicit network-side current state (an observation field the current harness does not yet provide, in which case the projector emits *unconfirmed*), and the night gate becomes a computed ledger (remaining dark hours $\times$ the dense-vs-sparse extra draw vs. reported SoC) that distinguishes ENERGY-INFEASIBLE from mere conservative advice. Replaying v4 and v5 over the same logged observations: v5 issues zero *unreachable* assertions without a network-side field (v4 fired at all five healthy-link false-alarm times; v5 labels them *unconfirmed*), and of the 528 dark decisions on which v4 printed a blanket “permanently kills” gate, the ledger classifies 189 as genuinely energy-infeasible and 339 as advisory—including every near-dawn decision (04:30–06:00), where the remaining dense extra draw is small and reported SoC suffices. This is a counterfactual mechanism witness, not an end-to-end v5 result.

Scope of the agent evidence. The live study is formative and single-model: it demonstrates the failure mode (access receipts as backhaul state), the direction of the v4 packet’s effect (fewer ineffective dense orders, fewer deaths, but confounded with format reliability and expert heuristics), and the need for the v5 statement discipline. It does not establish an independent certificate increment (the A1 packet also bundled day/night heuristics, geometric constants, confirmation rules, and note templates), does not wire L0 into the agent, and is not a powered service-significance test. The deterministic service, expiry, and safety results of RQ3/RQ4 do not depend on the LLM. The specified next validation—a matched-capability arm that receives the same constants and heuristics without the segment projector, v5 end-to-end runs with an explicit backhaul-state observation, and cross-model replication—is left as future work.

9 Related work

LLMs in emergency / network control. TopoLLM [7] uses an LLM with adaptive tool learning for post-disaster emergency network topology planning; subsequent lines let LLMs directly schedule UAV path/data collection and bandwidth, shape rewards for deep RL, or orchestrate infrastructure-recovery tools, including graph-guided candidate-plan repair and LLM-driven opera-

tion of real network-management APIs [8, 9]. These decide *network resources* or compose recovery tools; intent objects, budget/time constraints, candidate-plan checks, and local repair already appear in restoration systems, and LLM systems already orchestrate documented APIs [8, 9], so we do not claim “natural language to constraints” as novel. They do not maintain a long-horizon routine monitoring workflow across collection, configuration, return, and recovery under intermittent links and tight energy, and do not maintain an unknown-aware, three-class obligation state. Our delta is the per-obligation chain projection with an explicit unknown, the conditional structural statement, and the local-certainty release principle.

Scheduling, freshness, energy. An LLM in-context data-collection scheduler for UAV-assisted sensor networks supplies the closest state-model precedent—per-sensor bounded queues, exogenous arrivals, selectable objects under finite per-step service, battery/channel state, and a verifier feedback loop [10]; AoI and energy-harvesting link optimisation, including instantaneous-battery threshold policies and partially-observed energy-harvesting control [11, 12], and overbooking/scheduled-aware forwarding for space links [13] address pieces of the resource problem. Joint sampling/transmission/energy optimisation is well established and we do not claim it; our distinction is remote *persistent* configuration with Class A apply latency, the separated access vs. primary/backup return process, and—decisively—what the controller should *say and lawfully do* under partial observability when the mission is unfulfillable. Consistent network update work has long studied per-update validity, dependencies, and loop-freedom during transitions [14]; we rely on that lesson (versioning, dependencies, local repair are not new). Network calculus and DTN triage under extreme contacts are complementary deterministic characterisations of the walls we measure [15].

Packet discard, freshness, and DTN expiry. Discarding stale or expired packets is well established: AoI packet management pre-empts or drops old updates [4, 16, 17]; deadline-aware buffer discard has been coupled with adaptive ARQ for video [18]; a sensor transport suppresses generation when estimated delay exceeds a limit [19]; DTN bundle protocols attach a creation-time lifetime and expire bundles, alongside TTL/ encounter-based drop policies [5, 20, 6]; LoRa-LEO rural IoT systems instead schedule around estimated satellite contacts [21]. Our source-local disposition *is* standard per-record expiration and we prove the equivalence; we claim not the act of dropping but (i) setting the lifetime from the obligation’s end-to-end business deadline rather than packet age, a local buffer deadline, an estimated single-packet delay, or a bundle TTL; (ii) requiring disposition to be consistent across the access and backhaul segments, because withholding an expired backhaul send while the acknowledgement-driven FIFO repair cache keeps the record back-pressures fresh access—a cross-segment interaction absent from single-hop AoI discard and hop-local bundle expiry; and (iii) retaining every still-rescuable record, which AoI latest-only (whose objective ignores coverage) does not. DTN custody transfer [22] moves retention responsibility as a *forward* hand-off to decouple reliability from any end-to-end acknowledgement, keyed to custody rather than an obligation deadline, and has been migrated out of the BPv7 core [6]; it is an alternative architecture for the responsibility transfer we measure at the source, and a deployment study of custody hand-off versus source expiry is complementary future work.

10 Discussion, limitations, and threats to validity

Scope of claims. The throughput negative result is established at one main operating point against symmetric relaxations and strong (omniscient, MPC, greedy) baselines, and is scoped to the policy families tested; we make no global optimality or zero-remainder claim, and a better formulated controller could exist. The conditional S_{time} certificate is geometric and seed-invariant but presupposes a network-side unavailable interval; without it the statement is only conditional.

Online arrival attribution is deliberately partial (53.3%, exact on the labelled subset); treating its unknowns as a known segment would recreate the exact error we correct, and a deployment that can supply current (not forecast) backhaul state to the centre can shrink the unknown class—the v5 projector already consumes such a field.

Agent evidence. The live study uses one model with three outage seeds per arm (plus two healthy controls), each a single temperature-0 trajectory; per-run service variance is large at $n = 3$, and the arm contrast confounds the segment projector with expert heuristics, geometric constants, note templates, and output-format reliability (parse-failure rates differ 5-fold). The hardened harness (per-request accounting, symmetric scoring over every decision and both groups, the v5 unknown-aware projector and energy ledger) is in the repository so that the specified matched-capability, v5 end-to-end, and cross-model studies can be run without re-litigating the instrumentation; those runs are explicitly not claimed here. The live agent loop was not wired to the L0 floor, so deterministic RQ4 zeros are not transcribed to agent deaths.

Expiry assumptions. The central acknowledgement is modelled synchronously and without airtime, a pre-existing simplification that favours every queue discipline; the source-local expiry needs no downlink and is our primary mechanism, while gateway acknowledgements would consume Class A receive windows in deployment. During the outage the node tests deadlines against the public pre-change period $P=600$; the post-upgrade $P=300$ deadline enters only post-hoc accounting, never as an online input. The equivalence of obligation-keyed and deadline-keyed release holds for a single fixed-period mission; multi-obligation or mission-version overlap, where one record remains valid for a later obligation, requires release keyed to the record’s latest still-valid deadline and is left future work, as is a custody-transfer deployment comparison. Expired records forgo retrospective value after their deadline; where that matters they can be retained at the gateway rather than sent.

Model and provenance. The brownout model is absorbing; a restart sensitivity could move the energy wall but not the slot/capacity walls. Wider deployment ranges (backup rate/payload, node count, capacity, cloud cover, restart threshold) are not swept, and sub-point mean-service differences are not treated as significant. Centre knowledge of the outage interval assumes network-side operations visibility of the *current* state, not its duration; a conservative variant counts only deterministic backup slots. Standards text for one channel was obtained from a document-sharing host (top-grade document, lower-grade copy chain) because the official host was unreachable by TLS from our environment; clause numbers were cross-checked from two independent copies, and the international LoRa-plus-satellite and some vendor lines were not verified and are not used as fact. We did not execute cited systems’ reference implementations and claim no replication of them; one field-study endpoint was rate-limited from our network and is cited without its numbers.

Generality. The obligation chain, the three-class statement discipline with an explicit unknown, and the local-certainty release principle transfer to any intermittent, energy-limited, authority-bounded monitoring workflow; the specific walls and periods are parameterised by the deployment and must be re-derived per deployment.

11 Conclusion

For pre-disaster geohazard monitoring under intermittent connectivity, strong ordinary controllers already extract essentially all the on-time delivery that fixed resources allow over the policies we tested; the failures that remain are physical and, offline, cleanly attributable. The defensible role of an agent runtime is therefore neither to out-pack the network nor to drive safety-critical sampling, but to sit above a layered, execution-grounded obligation runtime. We separate what the field can know from what it cannot: at gateway arrival less than half of missed upgraded obligations admit

an exact segment label from local evidence and the rest must remain unknown, while complete attribution is an offline capability. That partial observability does not paralyse the edge—a node holds certain knowledge of its own records’ business deadlines and can stop worthless retransmission and night-time discharge without diagnosing the remote cause. This deadline-aligned expiry is a standard primitive whose value lies in its business-deadline basis, its zero-downlink source placement, and its cross-segment consistency, doubling outage on-time delivery while removing expired sends and deaths where gateway-only suppression merely moves cost to the access segment. Safety-critical constraints sit at the node because only there the observe-decide-act delay beats the intervention window; the centre is confined to non-pre-provisioned authorised changes and to statements that never upgrade unconfirmed to unreachable. A formative real-agent study confirms the interface failure this design targets and motivates the unknown-aware projector, while the deterministic guarantees rest on rules any planner can use. The result is a runtime that promises only what it can execute, says unknown when it cannot tell, and still acts—locally and correctly—on the things it knows for certain.

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